MICHAEL URGILEZ

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EDUCATION

Columbia University in the City of New York May 2027
Fu Foundation School of Engineering and Applied Science
Bachelor of Science in Mechanical Engineering – *Cumulative GPA:* 3.51/4.0 – Dean's List (Spring 2024, Spring 2025, Spring 2026)
- *Relevant Coursework:* Machine Design, Heat Transfer, Kinematics of Machines and Robots, Mechanics of Solids, Foundations of Data Science
- *Honors and Awards:* Columbia Engineering Lowey Family Foundation Scholarship (2024, 2025, 2026), Hispanic Scholarship Fund Scholar (2024, 2025, 2026)

SKILLS

- ***CAD & Manufacturing:*** NX, SolidWorks (Portfolio), Fusion 360, Engineering Drawing, GD&T, CNC Machining, 3D Printing, Laser Cutting, Soldering
- ***Analysis:*** Computational Fluid Dynamics (ANSYS), Finite Element Analysis, Circuit Simulation (LTspice)
- ***Programming & Tools:*** Python (data analysis, computer vision), Java, HTML, Git/GitHub, GitHub Copilot, LaTeX, GIS
- ***Languages:*** Spanish (Business Fluency), French (Proficient)

PROFESSIONAL EXPERIENCE

Engines Design Engineering Intern May 2026 – August 2026
GE Aerospace
- Cut GE9X fuel tip defects by 40% by leading a root-cause investigation, using data analysis and GD&T interpretation to identify drivers and ideate fixes
- Reduced combustor borescope inspection time by 50% by developing a MobileNetV2 computer vision tool that automates nozzle-level defect detection, winning 1st place at GE's inaugural Combustor & Hot-Frames AI Shark Tank
- Automated capability reporting with an app that generates reports from uploaded measurement data, plus an interactive HTML capability analysis
- Identified design and process improvements for combustor dome cost reduction that enabled a 20% labor-hour reduction
- Assessed combustor mixer nonconformance risk, applying GD&T to evaluate part deviations and substantiate dispositions that limit quality escapes
- Supported GE9X on-engine inspections of combustor hardware and completed durability mapping documentation to improve traceability

Research Assistant August 2025 – October 2026
Columbia University BioMEMS Laboratory – PI: Qiao Lin
- Fabricate polymer-enhanced microfluidic devices for biomolecule control and sensing, increasing graphene bond strength by 15%
- Analyze biological samples using MEMS-based affinity biosensors to characterize temperature-dependent biomolecular behavior
- Develop microfluidic models for compliance-based control devices, supporting the design of purification systems that use thermally controlled binding

Summer Intern June 2024 – August 2024
Paul M. Angell Family Foundation
- Led a GIS mapping project to identify high-impact areas for grantee partnerships across the US, optimizing resource allocation for the Social Impact Team
- Analyzed metrics for 139 grantee partner organizations in grant management systems, validating data in Excel and Python to inform data-driven decisions

PROJECTS

Omnidroid Walking Robot – Robotics Studio September 2026 – Present
- Designing a four-legged walking robot with a retractable sensor pod, actuated by serial-bus servos and controlled by a single-board computer

Scale Racing Car Casting and Molding – Materials and Processes in Manufacturing Spring 2026
- Modeled a toy-scale 1930s racing car in SolidWorks as part of a team, translating the design into molds for two production processes
- Designed the body mold and produced car bodies by silicone casting
- Programmed Fusion 360 CAM toolpaths to machine wheel mold cavities, then produced the wheels by plastic injection molding

Dual Axis Solar Tracker Spring 2026
- Designed and built a dual-axis solar tracker with Arduino, photoresistors, and servos to orient a PV panel toward peak irradiance
- Implemented closed-loop feedback control comparing light-sensor readings to adjust azimuth and elevation in real time
- Reduced travel time to the source of irradiance by 30% through 3D-printed photoresistor isolators and programmed environment calibration

Tesla Valve CFD Analysis – Fluid Dynamics November 2025 – December 2025
- Performed 15+ laminar CFD simulations in ANSYS (Re = 50–350) to quantify pressure loss and Tesla-valve diodicity up to 1.28
- Analyzed 1–5 repeating valve elements, identifying a linear diodicity increase of ~0.039 per element driven by geometric flow asymmetry
- Designed and validated insert geometry modifications, increasing diodicity by 8.65% at low Reynolds numbers

AQUAS Mission – Columbia University Robotics Club December 2024 – December 2025
- Minimized sensor buoy weight by 20% using assembly analysis in Fusion 360
- Designed a water collection device in Fusion 360 and repositioned internal electronics to improve stability, protecting them from water
- Prototyped the casing and electronics enclosure for an autonomous surface vehicle that treats algae blooms, using soldering, laser cutting, and CNC machining

LEADERSHIP & INVOLVEMENT

Co-President May 2026 – Present
Columbia MechE Mentors
- Lead a peer mentoring program pairing Mechanical Engineering third and fourth year students
- Expanded the program to all underclassmen and graduate students, growing participation by 50%

Vice President April 2025 – Present
American Society of Mechanical Engineers, Columbia University Chapter
- Plan and coordinate class-wide events, including dinners, networking events, and technical elective workshops
- Assess input from the Class of 2027 to present at weekly Executive Board meetings and monthly meetings with Mechanical Engineering faculty

Technology Chair August 2025 – Present
Society of Hispanic Professional Engineers, Columbia University Chapter
- Manage chapter digital infrastructure, including website management with GitHub and VS Code and document storage on Google Drive and Notion
- Designed a WhatsApp community with announcement and general channels, establishing a direct line of communication between E-Board and general body

Academic Vice President August 2025 – September 2026
Society for the Advancement of Chicanx/Hispanics and Native Americans in Science (SACNAS+), Columbia University Chapter
- Planned flagship programming (Graduate School Series, research pathway talks, résumé workshops) and oversaw the Big & Little mentoring program

Other Activities: Micro-G Mission Member – *Columbia Space Initiative*; Member – *Ecuadorians at Columbia University*, *Hispanic Pre-Law Society*